



Comparative Analysis of Soil Moisture Interpolation Techniques in Apple Orchards of Trentino Region

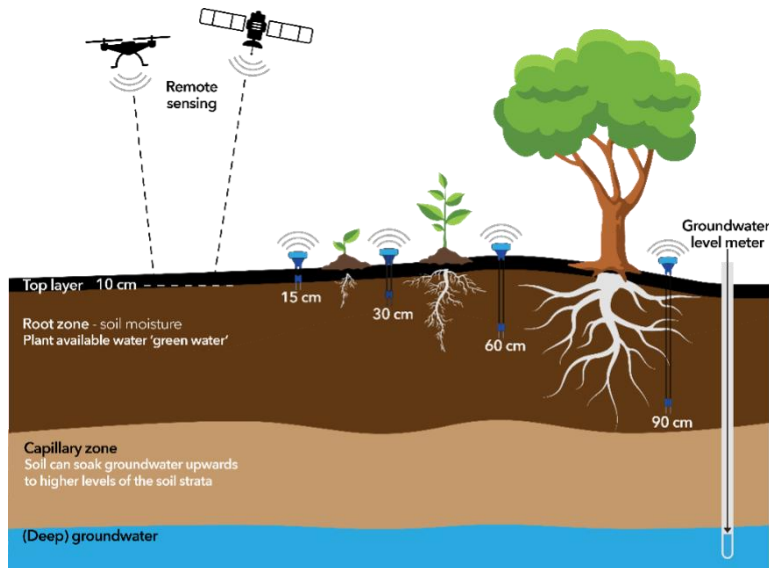
IRRITRE: geographical information system for precision irrigation in Trentino



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Introduction

The Importance of Soil Moisture



- Efficient monitoring and management of **soil moisture** are essential for determining optimal irrigation schedules, reducing water waste, and promoting plant health
- In precision agriculture soil moisture is measured by a **network of tensiometers** deployed in the fields
- Remote and IoT sensors can be combined to provide **intelligent decision support systems** for farmers

Figure: Diagram showing Remote and IoT sensors measuring field characteristics

Introduction

Why Soil Moisture Interpolation?

" Spatial interpolation of soil moisture is essential for transforming point-based measurements into continuous maps, enabling comprehensive analysis across large agricultural areas "

Study Overview

Dataset

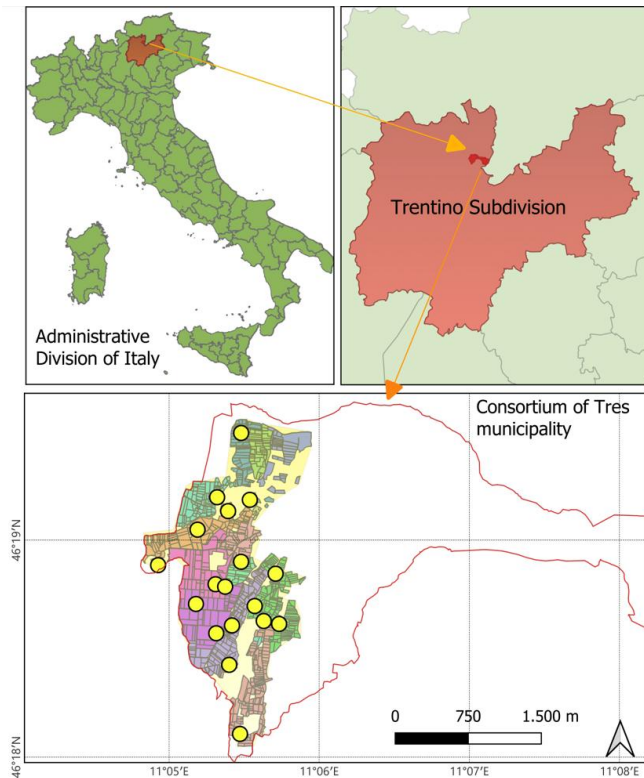
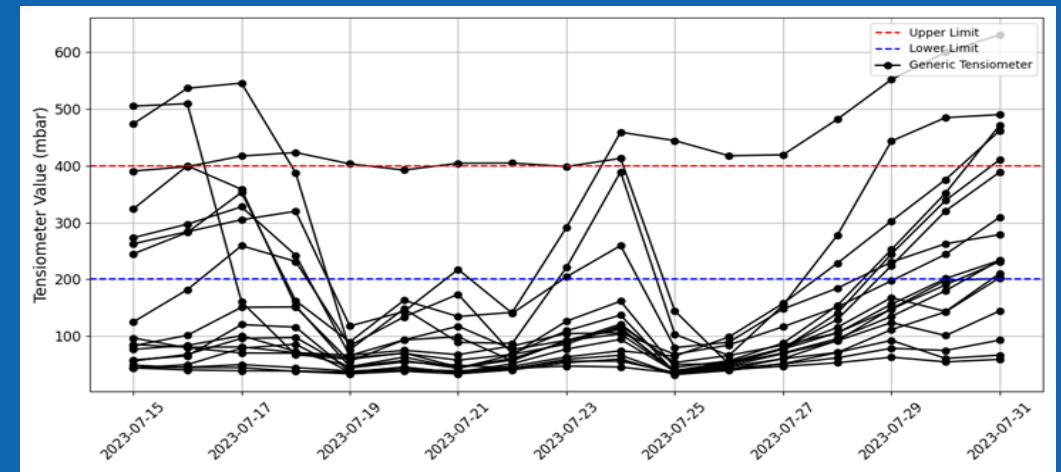


Figure: Map showing the geographical area of the Tres consortium, where the tensiometers are installed



Location: Tres consortium of Trentino, Italy

Period: July 15 to July 31, 2023

Data: collected from 18 tensiometers, installed at depth of 30 centimeters, with measurements taken every 15 minutes

Study Overview

Research Objectives



Applicability

Apply spatial interpolation methods to create maps for specific moments



Comparison

Evaluate the performance of the methods using a reliable and consistent methodology



Validation

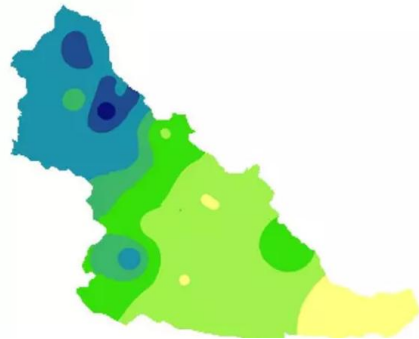
Statistically assess whether the differences between methods are significant

Inverse Distance Weighting (IDW)

$$\hat{f}_{IDW}(y) = \frac{\sum_{i=1}^N \frac{f(x_i)}{d(y, x_i)^p}}{\sum_{i=1}^N \frac{1}{d(y, x_i)^p}}$$

where $d(y, x_i)$ is the distance between point y and point x_i , and p is the power parameter that controls the weights of the distances.

Example Map:

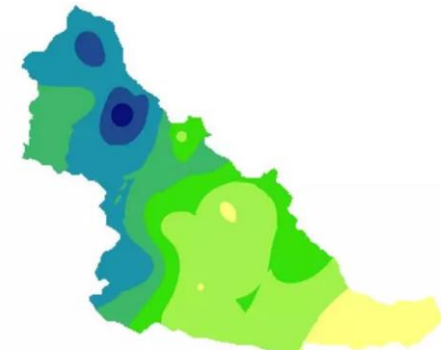


Ordinary Kriging (OK)

$$\hat{f}_{OK}(y) = \sum_{i=1}^N \lambda_i f(x_i)$$

where λ_i are the Kriging weights assigned to the known points x_i ; these parameters are determined by solving a system of linear equations based on the variogram.

Example Map:



Methodology

How to evaluate?

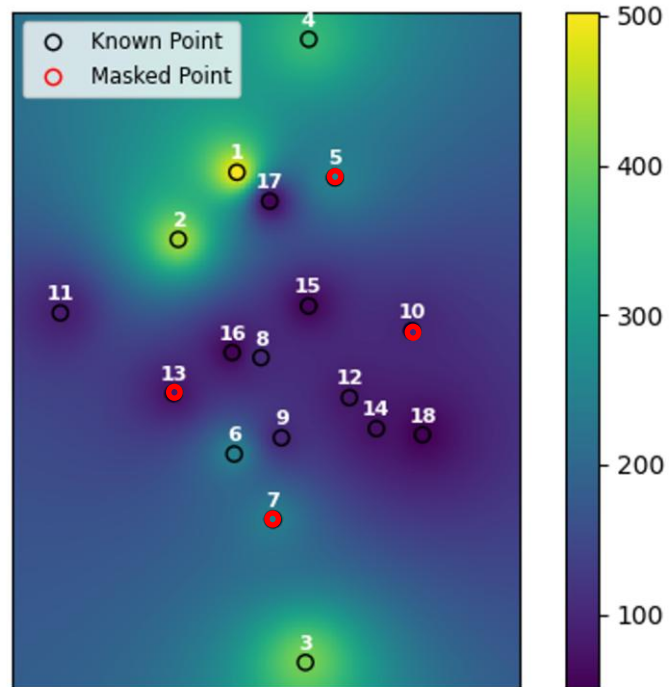


Figure: Interpolation map built with IDW on July 15, 2023 at midnight

- The data points need to be split into known and unknown categories to accurately evaluate estimation error and ensure reliable results
- RMSE serves as the primary evaluation metric because it effectively captures the differences between estimated and observed values
- Validation techniques and statistical methods are essential for validating the robustness of the results and preventing overfitting

Results

First Comparison

Graphical comparison of daily average RMSE values between IDW and OK interpolation methods, calculated using an 80-20 holdout validation:

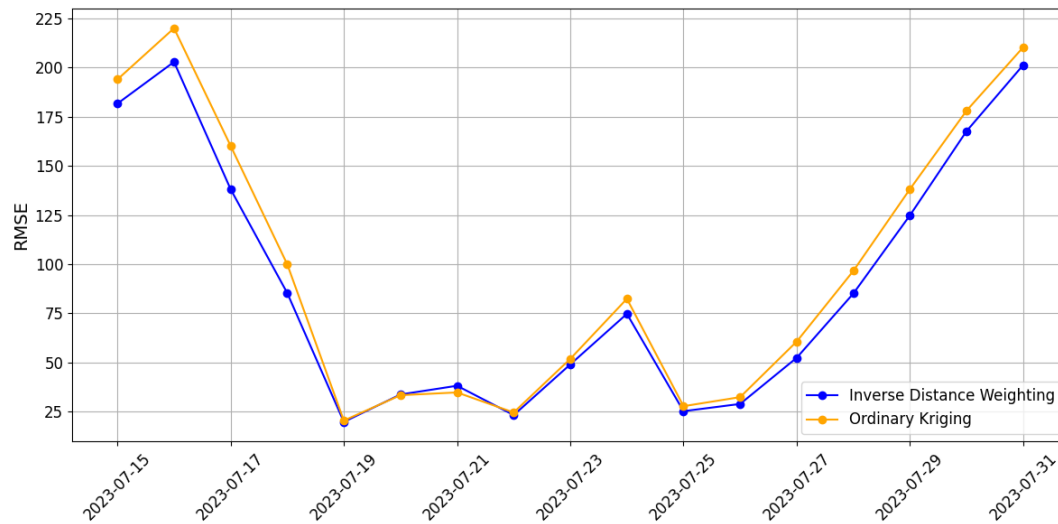


Figure: Comparison of average RMSE values of Inverse Distance Weighting and Ordinary Kriging over time

- RMSE values show significant variation depending on the time period
- IDW slightly outperforms OK in certain periods, with near-equivalent performance in others
- Results are highly dependent on the validation method (which points are used for interpolation)

Results

Cross-Validation Results

Graphical cross-validated comparison of RMSE values between IDW and OK interpolation methods:

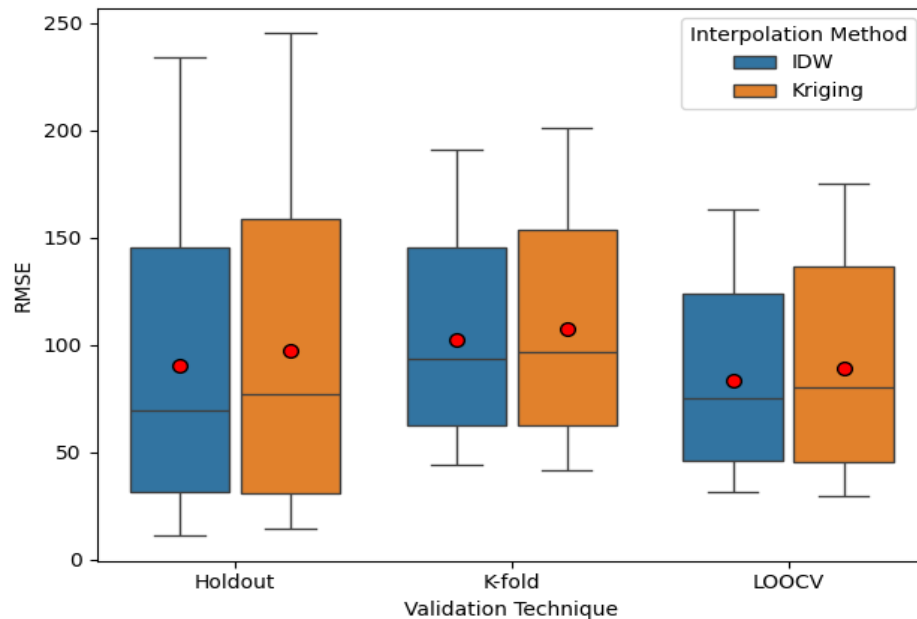


Figure: Comparison of RMSE distributions for Inverse Distance Weighting and Ordinary Kriging across various Validation Techniques

	IDW	OK	IDW – OK
Holdout	90,13	97,43	-7,30
K-Fold	102,60	107,28	-4,68
LOOCV	83,58	89,06	-5,48

- **IDW consistently outperforms OK across all validation techniques**
- **The differences between the two methods are minimal, raising questions about their statistical significance**

Results

Statistical Analysis

01

Compute the Differences

Calculate the RMSE differences between IDW and OK across all timestamps

02

Verify Assumptions

Use Kolmogorov-Smirnov test to check the normality assumption of the RMSE differences distribution

03

Apply Paired t-Test

Conduct a paired t-test to assess the statistical significance of the observed differences

	K-S Test (p -value)	Paired t-Test (p -value)	Result
Holdout	0.96 (0.088)	-17.92 (≈ 0)	✓
K-Fold	0.96 (0.074)	-9.88 (≈ 0)	✓
LOOCV	0.97 (0.208)	-11.49 (≈ 0)	✓

- With a 5% significance level (p -value < 0.05), the K-S test slightly fails to reject the null hypothesis, indicating that the assumption of normal distribution is satisfied
- Conversely, the paired t-test consistently rejects the null hypothesis, demonstrating that the RMSE differences are statistically significant

Conclusion

Key Takeaways

01

In this study, IDW outperforms OK for soil moisture estimation. IDW is effective with sparse sensor data, whereas OK relies on a denser network to fully leverage spatial correlations.

Performance

02

IDW's simplicity and low computational cost make it a practical solution for real-time irrigation management under current field conditions. The error results are relatively small, given the context of the actual case study.

Applicability

03

The performance may vary with different datasets or larger sensor networks. These results are specific to this region and sensor setup, and IDW's smoothing of spatial variations could limit its accuracy in more complex environments.

Limitations

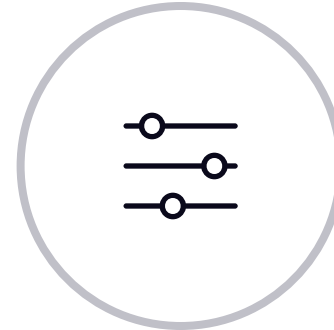
Conclusion

Future Directions



Sensors

Expand the sensor network density to maximize the potential of both methods and improve spatial resolution



Methods

Evaluate other interpolation techniques (splines, co-kriging, deep learning) to identify potential improvements



Variables

Integrate other correlated factors, such as irrigations, to enhance the accuracy and relevance of the analysis



Scalability

Investigate how the methods perform in other regions to evaluate the scalability and to diverse conditions

「thank you」